Triangulation Collapse: How LLM-Mediated Search Removes the User's Last Line of Defense in High-Stakes Queries

Mohib Fatima
Independent Researcher | mohib.fatima.edu@gmail.com
Technical Report · June 2026 · Revised November 2026

Abstract

For decades, search engines functioned as triangulation tools. A user searching for information about a worrying symptom, an employment dispute, or a financial decision would encounter multiple sources, compare them, disagree with some, and arrive at a conclusion of their own making. That comparison process (imperfect, slow, occasionally frustrating) was also the primary error-correction mechanism available to non-expert users navigating high-stakes information domains. The rapid adoption of LLM-mediated search, which synthesizes ranked results into single authoritative-sounding responses, may be eliminating this behavior. I call this elimination Triangulation Collapse (TC). This paper characterizes the structural conditions under which TC would arise. It gives the engagement-accuracy decoupling mechanism a minimal formal model with three propositions, each of which holds under stated assumptions, and it reports a direct test of one of them, Synthesis Dilution, against a live model: 288 outputs across four domains and two task formats, scored by a judge that I checked against my own blind labels. No output endorsed the overconfident claim, and I report this as a null result for the model tested. I then discuss four high-stakes domains (health, legal, financial, and hiring) as hypotheses and not as findings, and propose SARI (Stakes-Aware Re-ranking and Inference), a design for re-introducing accuracy-oriented signals into LLM search pipelines for queries where the cost of error is high. SARI is not implemented or evaluated. The core argument is that LLM search systems may have inherited the engagement-optimization logic of their underlying ranking infrastructure, and for high-stakes queries, engagement and accuracy are not the same thing. Such a mismatch would not show up in standard quality metrics, and Proposition 3 shows why for the single-answer interface. Whether the mismatch is large in deployed systems is an empirical question this paper opens and does not close.

1. Introduction

There is a specific kind of query that search engines have always handled poorly. Not the easy ones: restaurant recommendations, sports scores, weather. The hard ones. What do these symptoms mean? Do I have a case against my employer? Should I be worried about this clause in my loan agreement? Can this company reject my application for this reason?

For these queries, the right answer is almost never simple. It depends on jurisdiction, individual circumstance, the severity of the specific situation, and often on details the user hasn't thought to include in a six-word search query. The reasonable response to this kind of question is: here are several perspectives, here are the authoritative sources, here are the caveats, and here is where you should go for professional guidance.

That is not what LLM-mediated search systems are built to deliver. They are built to deliver one answer, confidently, immediately, in a format that feels complete. In industry practice, ranking and product decisions are commonly guided by engagement signals such as clicks, dwell time, return visits, and query reformulation rates (Section 2.1). Content that performs well on those signals is not always accurate; it is content that is compelling. In most domains, that's fine. In high-stakes domains, it's a problem that is rarely measured.

Broad surveys of foundation model risk have catalogued failure modes ranging from bias amplification to homogenization of downstream applications (Bommasani et al., 2021), but the specific behavioral consequence I want to describe here sits outside that catalogue. Before LLM synthesis became the default interface, a user who received an unsatisfying or uncertain answer from one source would click another link. Then another. They would triangulate, building a picture from multiple sources, noticing where they agreed and where they diverged, and forming a judgment. That process was slow and imperfect. But it was also protective. The user was the last line of defense against any single source being wrong.

That behavior may now be designed away. A stated goal of LLM-mediated search is to reduce the need to click through multiple links. Whether users have in fact stopped cross-referencing is a behavioral question this paper does not measure. I treat it as the premise that the formal model makes precise and that future work should test. The output presents as authoritative and complete; whether it is accurate is a separate question that the interface no longer prompts the user to ask.

This paper names and formalizes this shift as Triangulation Collapse, documents the engagement-accuracy decoupling mechanism that drives it, including a minimal formal model in Section 3.3 that gives each stage of the mechanism a proposition and a short proof, in the same spirit as the formal treatment I give Feedback-Induced Distribution Shift (Fatima, 2026), reports a small live-model test of one of its propositions (Section 6), traces the hypothesized consequences across four high-stakes domains, and proposes a concrete mitigation framework. The central claim is not that LLM search is bad. It is that LLM search may have inherited an optimization target (user engagement) that was designed for low-stakes queries and scales poorly to high-stakes ones. That mismatch would have consequences. And no standard evaluation framework I am aware of is designed to detect them; Section 3.3 shows that, for the single-answer interface, this blindness is not an oversight but a structural property of how single-answer synthesis is presented.

2. Background and Related Work

2.1 How Search Ranking Optimization Works

Modern search ranking systems use a combination of signals to determine which pages surface for a given query. The precise mix is proprietary, but the signals that drive optimization decisions are well-understood from industry practice: click-through rates, dwell time (how long a user spends on a page before returning to search), bounce rates, return visit frequency, and query reformulation (whether a user searches again immediately after visiting a result, suggesting dissatisfaction). These signals are practical proxies for relevance. If users click a result and stay, the result was probably useful. If they leave immediately, probably not.

The problem is that these proxies are not neutral. They measure whether content is engaging, not whether it is accurate. For most query types, this distinction doesn’t matter much. For high-stakes queries, it matters enormously. A health article that lists many possible causes, including rare and serious ones, would be expected to generate more dwell time than one that says ‘these symptoms usually resolve on their own,’ based on the engagement mechanics described above. The same logic applies to legal content: an article implying a strong case would be expected to generate more return visits than one that carefully explains why most employment claims are difficult to prove. Engagement-optimized ranking, applied to high-stakes content, would be expected to reward alarm and oversimplification. These are assumptions of the model in Section 3.3, supported here by argument and by the mechanics of ranking, not measured in this paper.

2.2 LLM Synthesis and Source Inheritance

LLM-mediated search systems (AI Overviews, conversational search interfaces, retrieval-augmented generation pipelines) typically operate by retrieving a set of candidate documents from a ranked index, then synthesizing those documents into a single generated response. The synthesis step is where LLM capabilities are most visible. But the retrieval step, the set of documents that gets synthesized, is almost always determined by the same engagement-optimized ranking infrastructure that underlies traditional search.

This means LLM search systems inherit the biases of their source rankings, a source-level analogue of the model-output feedback loops documented in training data (Taori & Hashimoto, 2023), except here the loop runs through retrieval rather than retraining. If engagement-optimized ranking tends to surface alarming health content, the LLM synthesis of that content will tend toward alarming health conclusions. The synthesis adds an additional layer of concern: it renders the source documents invisible. In traditional search, a user could look at the URL, assess the source, and decide whether to trust it. In LLM synthesis, the sources are either hidden or listed in a collapsed reference section that users may not expand. The content has been laundered through the model's voice, and that voice sounds authoritative regardless of what it's synthesizing.

2.3 Information Foraging and Triangulation Behavior

Information foraging theory (Pirolli & Card, 1999) describes how users navigate information environments using information scent: cues that indicate whether a source is likely to contain what they need. In traditional multi-source search, users iteratively follow scent signals, compare what they find across sources, and refine their understanding. This process is not just about finding an answer: it is a form of epistemic quality control. Encountering multiple sources that agree increases confidence. Encountering disagreement prompts deeper investigation. A user who finds three sources saying different things about a symptom's significance would plausibly be more likely to consult a doctor than one who receives a single confident AI summary, a connection the formal model in Section 3.3 makes precise.

LLM-mediated search collapses this foraging process into a single synthetic output. The user receives what appears to be the result of comprehensive research, conducted and synthesized by an authoritative system. The behavioral consequence (reduced verification, reduced source diversity, increased acceptance of the first answer received) is what I am calling Triangulation Collapse. Section 3.3 gives this consequence a formal statement: it treats the user's verification decision as a function of observed disagreement across sources, and shows that single-answer synthesis fixes that function's input at its minimum value regardless of how much the underlying sources actually disagreed.

2.4 Prior Work on Search Bias and LLM Accuracy

The tension between engagement and accuracy in search has been discussed qualitatively in media criticism and public health research for years (Eysenbach & Köhler, 2002). Academic treatment of bias in search and language technology has focused primarily on political and demographic representation (Noble, 2018; Bolukbasi et al., 2016), and separately on subgroup performance disparities under aggregate optimization (Hashimoto et al., 2018; Fatima, 2026), rather than the accuracy-stakes dimension I address here. Work on LLM hallucination (Maynez et al., 2020; Mallen et al., 2022) documents the phenomenon of factually incorrect generation but does not address the specific amplification mechanism introduced when LLMs synthesize engagement-optimized ranked sources. Liang et al.'s HELM framework (2022) provides the most rigorous evaluation infrastructure for LLM capabilities but measures point-in-time accuracy rather than the longitudinal trust and behavior changes that Triangulation Collapse describes. To my knowledge, no prior work has named or formally characterized the elimination of multi-

source verification behavior as a distinct AI deployment failure mode, though the empirical work in Section 2.5 measures part of the same problem from the output side.

2.5 Verifiability, Trust, and Manipulation in Generative Search

Two recent lines of work are close to this paper. Liu et al. (2023) audited four commercial generative search engines on 1,450 queries and found that roughly half of generated statements lacked supporting citations and roughly three quarters of citations supported the statement they were attached to; responses that sounded most convincing were often the least well supported. Chen et al. (2026) introduce SearchGEO, which measures how often LLM search agents turn attacker-published web content into endorsed recommendations; across 13 backends, attack success ranged from 0.0% to 31.4%.

Both measure properties of the output or the backend. TC is about the user: how a single-answer interface changes what the user does next, even when sources are faithfully synthesized and nobody is attacking. The two are complementary. Unsupported statements make the loss of triangulation costlier, and adversarial manipulation is one way a biased source set can enter the synthesis context C in Section 3.3. What this paper adds is a formal account of the behavioral step and a test of one of its premises.

3. Triangulation Collapse: Definition and Mechanism

3.1 Definition

Triangulation Collapse (TC) is the systematic elimination of multi-source verification behavior in users of LLM-mediated search systems, driven by interface design that presents synthesized responses as authoritative single answers, and resulting in amplified propagation of engagement-accuracy bias in high-stakes query domains.

TC is not a failure of the LLM itself. The model may be performing exactly as designed, generating fluent, confident, coherent responses. TC is a systems-level failure: the combined effect of engagement-optimized ranking, invisible sourcing, and authoritative-sounding synthesis on user behavior in contexts where that behavior change carries real-world consequences.

It is also worth being specific about what TC is not. It is not simply the problem of LLM hallucination: the model generating content that doesn’t appear in any source. TC describes what happens even when the model faithfully synthesizes its sources, if those sources are themselves biased by engagement optimization. The model can be technically accurate relative to its inputs and still produce systematically misleading outputs. That distinction matters for both research and for mitigation design, and it is exactly what the formal model in Section 3.3 is built to isolate: each proposition holds even under the assumption of perfectly faithful synthesis.

3.2 The Engagement-Accuracy Decoupling Mechanism

The mechanism driving TC operates in three stages. I’ll describe each stage here in operational terms and give each one a formal counterpart in Section 3.3.

Stage 1: Engagement-Biased Source Selection. Ranking systems surface content with high engagement signals. In high-stakes domains, the content that performs best on engagement metrics tends to share specific characteristics: it addresses the user’s concern directly and emotionally, it implies a clear (often alarming) conclusion, it uses accessible language, and it is structured for scanability rather than nuance. Clinical guidelines, academic papers, and jurisdiction-specific legal resources, which tend to be accurate but hedged, dense, and careful, perform poorly on engagement metrics. They are frequently outranked by consumer health articles, legal advice blogs, and financial commentary sites that optimize for emotional resonance.

Stage 2: Synthesis Without Source Transparency. An LLM receives the top-ranked documents as context and generates a synthesized response. The synthesis is optimized for coherence and fluency, not for representing the diversity of source quality. A set of documents that includes one high-quality authoritative source and three engagement-optimized sources would be expected to produce a synthesis that reflects the dominant pattern across sources, which, for high-stakes topics, is the engagement-biased majority. The authoritative source's careful hedging may be diluted. The confident, alarming, or oversimplified claims from the engagement-optimized majority may be amplified. Section 6 tests this directly.

Stage 3: Behavioral Lock-In Through Interface Design. The synthesized response is presented in a format that signals completeness: a structured paragraph or bulleted list with an authoritative tone, often with a collapsed source list. The interface communicates, whether intentionally or not, that the research has been done. The user's behavioral instinct to check another source is plausibly suppressed. And since the cost of that verification was already low (clicking another link), its elimination is nearly invisible. No individual user notices that they've stopped triangulating. But in aggregate, across millions of high-stakes queries, the consequences could accumulate.

3.3 A Formal Model of the Decoupling Mechanism

The three stages above are operational descriptions. Each one has a minimal formal counterpart, elementary in the same sense as the FIDS formalization (Fatima, 2026): a dominance argument, a weighted average, and a degenerate function input. The goal is the same in both papers: to turn a qualitative description into a claim that is true by construction under stated assumptions. That makes the mechanism precise. It does not show that the assumptions hold in practice, and Section 6 tests one that does not appear to.

Setup and notation. For a given high-stakes query, let the candidate document pool be $D = \{d_1, \ldots, d_m\}$. Each document d_j has an engagement score e_j (the ranking signal) and a reliability score $a_j \in [0,1]$ (the probability that a claim drawn from d_j is correct for the user's specific situation). Retrieval selects the top-k documents by e_j to form the synthesis context $C \subset D$. In Stage 2, each document in C receives a synthesis salience weight $\beta_j \geq 0$ reflecting how much it shapes the model's output (e.g., proportional to how often its claims recur across the context). In Stage 3, let $\delta_{obs} \in [0,1]$ denote the disagreement across sources as observed by the user, and let $p_{verify} = \varphi(\delta_{obs})$ be the probability the user seeks further verification, where φ is non-decreasing and $\varphi(0) = p_{floor} \geq 0$. The model leaves φ unspecified beyond monotonicity. I do not claim it matches measured human behavior; anchoring φ in user studies is future work.

Proposition 1 (Engagement-Accuracy Decoupling: Stage 1). Engagement-ranked source selection can achieve the maximum attainable expected engagement while the expected reliability of the retrieved set is arbitrarily worse than the average reliability available in the candidate pool, whenever engagement and reliability are inversely ordered among candidates.

Setup. Partition the candidate pool D into an accurate set A (mean reliability $\bar{a}_A$) and an engagement-optimized set G, and let $a_{max}^G = \max_{j \in G} a_j$. Suppose $a_{max}^G < \bar{a}_A$ and every document in G has strictly higher engagement score than every document in A, the empirical pattern described in Section 2.1. Retrieval selects the top-k documents by e_j, with $k \leq |G|$.

$$C = \textit{top-}k_j\ e_j \ \Rightarrow\ C \subseteq G \ \Rightarrow\ E[a \mid C] \leq a_{max}^G < \bar{a}_A$$

Proof. Because every element of G strictly dominates every element of A in engagement score, the k highest-engagement documents are drawn entirely from G regardless of $k \leq |G|$ or of the values a_j take within A. Every document in C therefore has reliability at most a_{max}^G, so $E[a \mid C] \leq a_{max}^G$, and the gap

satisfies $\bar{a}_A - E[a \mid C] \geq \bar{a}_A - a_{max}^G$. That bound can be made arbitrarily close to 1 by construction ($\bar{a}_A \to 1$, $a_{max}^G \to 0$) without changing C or its engagement rank. ■

Why this matters. The selection rule that produced C is, by definition, engagement-optimal: it cannot be improved on the metric it is optimizing. The reliability gap it leaves on the table is not a bug in the ranking algorithm; it is what an engagement-optimal rule does whenever engagement and reliability are inversely ordered, which Section 2.1 argues may be the norm for high-stakes queries. No amount of tuning the engagement signal closes this gap, because the gap is a property of the ordering, not the ranking method.

Proposition 2 (Synthesis Dilution: Stage 2). A single accurate, hedged source in a synthesis context contributes an arbitrarily small share of the synthesized output's effective position as the salience-weighted share of engagement-optimized sources grows, regardless of how important the hedge is.

Setup. Let the synthesis context contain one hedged source with salience β_h and claim position y_{hedge}, and $n-1$ confident, engagement-optimized sources each with salience β_c and claim position $y_{confident}$. Model the synthesized output's effective position as the salience-weighted average of source positions:

$$S_n = [\beta_h y_{hedge} + (n-1)\,\beta_c y_{confident}] / [\beta_h + (n-1)\,\beta_c]$$

Result. As $n \to \infty$ (or as $(n-1)\beta_c/\beta_h \to \infty$), $S_n \to y_{confident}$. The hedged source's influence on S_n is bounded by $\beta_h / (\beta_h + (n-1)\beta_c)$, which $\to 0$ independent of the semantic importance of y_{hedge}.

Proof. Direct limit of the weighted average as $(n-1)\beta_c$ dominates the denominator; the coefficient on y_{hedge} is $\beta_h/(\beta_h + (n-1)\beta_c) \to 0$, so $S_n \to y_{confident}$ regardless of the fixed value y_{hedge} takes. ■ This is a statement about a weighted-average model of synthesis. Whether a real LLM synthesizes this way is the question Section 6 tests.

Why this matters. This is structurally the same masking phenomenon shown for Feedback-Induced Distribution Shift's Aggregate Masking Effect (Fatima, 2026): a bounded quantity is attenuated below any threshold by increasing the weight of a majority, regardless of the magnitude of what is being attenuated. There, the weight was traffic share; here it is synthesis salience. The same proof technique applies because the same structure, a weighted average with a shrinking-weight minority term, governs both. The practical consequence is that including an authoritative source in the synthesis context, without weighting it, would not be sufficient to preserve its influence, if synthesis behaves as the model assumes: presence in C (Proposition 1's output) does not imply proportional representation in the output.

Proposition 3 (Verification Suppression: Stage 3). Presenting search results as a single synthesized answer fixes the user's observed source disagreement at its minimum possible value, provided the interface exposes no other channel through which disagreement is visible, which collapses the probability of seeking further verification to its floor rate, independent of how much the underlying sources actually disagreed.

Setup. Let $p_{verify} = \varphi(\delta_{obs})$ as defined above, with φ non-decreasing and $\varphi(0) = p_{floor}$. Let $\Delta_{true} \in [0,1]$ denote the true dispersion among the sources' underlying claims (e.g., the variance of $\{y_1, \ldots, y_n\}$ for the sources in C). In multi-source presentation, the user observes $\delta_{obs} = \Delta_{true}$ directly. In single-answer LLM synthesis, the interface presents one output and no other disagreement channel (for example expanded citations, alternative answers, or hedging that varies with the sources), so $\delta_{obs} \equiv 0$ by construction, regardless of the value of Δ_{true}.

$$\delta_{obs}^{synthesis} \equiv 0 \;\Rightarrow\; p_{verify} = \varphi(0) = p_{floor} \;\; \text{for all } \Delta_{true} \in [0,1]$$

Proof. By definition p_{verify} depends on δ_{obs} only through φ. Since the single-answer interface fixes δ_{obs} at 0 independent of Δ_{true} (under the no-other-channel condition), $\varphi(\delta_{obs}) = \varphi(0) = p_{floor}$ for every value Δ_{true} can take, including Δ_{true} close to 1 (maximal underlying disagreement). ■

Why this matters. This is the formal version of Section 3.4's claim that standard metrics miss TC. If satisfaction and verification-seeking are both functions of δ_{obs} rather than Δ_{true}, as the model assumes, and δ_{obs} is architecturally fixed at 0 by single-answer synthesis, it follows that no metric computed from user behavior at the interface (clicks, satisfaction, return visits) can respond to Δ_{true} through δ_{obs}, because the interface has already severed that dependency before the user acts. The invisibility described qualitatively below is therefore not a measurement gap that better instrumentation could close from the demand side, under these assumptions; it requires either restoring δ_{obs} as a function of Δ_{true} (which is what SARI's uncertainty-surfacing intervention in Section 5.4 does) or measuring Δ_{true} directly on the supply side.

3.4 Why Standard Metrics Miss TC

Triangulation Collapse would be invisible to almost every quality metric currently in use. Average accuracy on benchmark queries doesn't capture it because TC is a behavioral phenomenon, not a model accuracy phenomenon. User satisfaction scores don't capture it because users who receive confident synthesized answers would be expected to report high satisfaction: they feel informed, even when they aren't; by Proposition 3, satisfaction is a function of $\delta_{obs} \equiv 0$, not of Δ_{true}, so it cannot distinguish a synthesis of genuinely agreeing sources from a synthesis that papered over real disagreement. Click-through rates don't capture it because the whole point of TC is that users aren't clicking through. The only metrics that would detect TC are ones that track longitudinal health, legal, or financial outcomes for users who relied on AI search answers, and, to my knowledge, nobody is collecting those systematically.

This invisibility is not an accident, and Proposition 3 shows it is not incidental either under the model's assumptions: it is what the single-answer interface produces by construction. Systems optimized for engagement generate satisfied users in the short term. The cost of TC is paid downstream, by the user who delayed seeking care, filed a weak legal claim, made a poorly-informed financial decision, or accepted a hiring rejection that was potentially discriminatory. That cost doesn't appear in any product dashboard.

3.5 Relation to LLM-Based Evaluation

TC also bears on how high-stakes search answers are evaluated, including by LLM judges. A judge that sees only the final synthesized answer receives the same $\delta_{obs} \equiv 0$ view that the user does, so its score cannot reflect whether the underlying sources agreed. LLM judges also have documented biases toward surface features such as position and verbosity (Zheng et al., 2023), and I hypothesize, without testing it, that confident phrasing can inflate scores in a similar way.

Two implications follow if the hypothesis holds. First, a judge evaluating a high-stakes answer should be given the source set and asked whether disagreement among the sources is represented in the answer, which is a check of δ_{obs} against Δ_{true}. Second, any automatic scorer of hedging or confidence, including the judge I use in Section 6, needs validation against human ratings before its output is treated as a measurement. I check the one judge I use against my own labels on 60 outputs, and I do not otherwise test judges in this paper.

4. TC Across High-Stakes Domains

The four domains below are hypotheses derived from the model and from established patterns in the search-quality literature. They are not findings of this paper. I call them high-stakes under the working definition in Section 5.2; no standard definition exists, though the category resembles the "Your Money

or Your Life" pages that search quality guidelines treat with extra care (Google, Search Quality Evaluator Guidelines).

4.1 Health Queries

Health is a domain where TC would plausibly be both widespread and consequential. Symptom queries, the most common health searches, illustrate the mechanism cleanly. A user searching for the cause of a recurring headache, an irregular heartbeat, or unexplained fatigue will encounter a ranked landscape dominated by content that lists the broadest possible range of conditions, including rare and serious ones, because comprehensiveness and mild alarm drive engagement.

When an LLM synthesizes this landscape, the result is typically a paragraph that acknowledges multiple possible causes but emphasizes the ones that generate the most concern. The synthesis usually includes a disclaimer to consult a doctor. But the disclaimer is appended to a confident synthesis, and in practice it functions as a legal hedge rather than a behavioral prompt. By Proposition 3, users who receive a confident answer with a disclaimer appended have their observed disagreement fixed at zero regardless of the underlying uncertainty, which predicts lower verification-seeking than users who receive multiple genuinely uncertain sources that visibly disagree with each other.

The old triangulation process (clicking four different medical sites, noticing that two said 'usually benign' and two said 'potentially serious,' deciding to monitor the situation) created calibrated uncertainty. The LLM synthesis of those same four sites creates confident incompleteness. The user receives the alarming content of the engagement-optimized majority (Proposition 2, if synthesis behaves as modeled) without the protective uncertainty of having seen the sources disagree (Proposition 3).

4.2 Legal Queries

Legal queries present a distinct but related pattern. Jurisdiction-specific legal information is extremely difficult to surface correctly through engagement-optimized ranking, for a simple reason: accurate legal answers almost always depend on jurisdiction, specific facts, and procedural history: details that a short search query never contains. The content that performs best on engagement signals is content that implies general applicability and clear outcomes. 'Can my employer do this?' articles that say 'probably not, here's what you can do' would be expected to generate more engagement than articles that say 'it depends on your state, your contract, your classification, and several other factors,' following the same engagement dynamics described in Section 2.1.

LLM synthesis of this content can produce answers that sound specific and actionable but are often neither. A user asking about wrongful termination or housing discrimination receives a confident summary that implies a general legal standard, without the jurisdiction-specific, fact-specific hedging that would be present in any answer given by an actual attorney. The user may act on that answer, or decide not to act believing they have no case, based on a synthesis of engagement-optimized content that was never designed to be accurate for their specific situation.

4.3 Financial Queries

Financial queries expose the engagement-accuracy decoupling most starkly in the gap between investment and personal finance content. Content about investment opportunities, market predictions, and financial products that performs well on engagement metrics tends to be directional, confident, and action-oriented. 'Here's why this sector will outperform' would be expected to generate more engagement than 'here are the relevant risk factors to consider,' for the same reason. Financial advice that includes appropriate caveats and recommends professional consultation would be expected to generate less engagement than financial advice that sounds like insider knowledge, for the same reason.

An LLM synthesizing the top-ranked financial content for a query like ‘should I refinance my mortgage now’ or ‘is this a good time to buy’ will tend to produce an answer that reflects the confident, directional tone of its engagement-optimized source documents. The user, who previously might have read three financial sites, noticed their different predictions, and concluded that nobody actually knows, now receives a synthesized answer that sounds more certain than any individual source, because synthesis strips the hedging while retaining the directional content.

4.4 Hiring and Employment Queries

Hiring-related queries represent perhaps the most acute form of TC because the user is often in a situation of significant power asymmetry and limited legal literacy. A job applicant trying to understand whether a rejection or non-interview was potentially discriminatory, or an employee trying to understand their rights regarding a non-compete clause, is making a decision with real career consequences based on information they are likely accessing for the first time.

The engagement-optimized content landscape for hiring queries tends to favor confident, general claims about worker rights: content that is emotionally resonant but often jurisdiction-dependent and case-specific to the point of being misleading in individual situations. An LLM synthesis of this content may give a user false confidence that they have a legal claim when they don’t, or false hopelessness that they have no recourse when they do. In both cases, the synthesis is not hallucinating: it is faithfully representing the biased source landscape it was given. And the user has lost the triangulation process that might have prompted them to consult a professional.

5. SARI: Stakes-Aware Re-ranking and Inference

5.1 Framework Overview

The core problem TC describes is an architectural mismatch: LLM search systems apply a single optimization logic (engagement-based ranking) to a heterogeneous query space that includes both low-stakes queries (where engagement and accuracy correlate reasonably well) and high-stakes queries (where they don’t). SARI is a framework for correcting that mismatch by introducing query-stakes awareness at two points in the search pipeline: at re-ranking, before the LLM receives its source documents, and at inference, when the LLM generates its response.

SARI does not require replacing the underlying ranking infrastructure. It operates as a modulation layer that activates on high-stakes query detection and adjusts both source selection and generation behavior accordingly. This is an important design choice: a framework that requires replacing ranking infrastructure is not deployable in practice. A framework that adds a stakes-aware layer on top of existing infrastructure can be.

5.2 Stakes Query Classification

The first component of SARI is a stakes classifier: a lightweight model trained to identify queries that fall into high-stakes domains. I define a high-stakes query as one where: (1) the user’s decision or action based on the answer may have significant personal consequences (health, legal rights, financial outcomes, employment), and (2) the accuracy of the answer depends on individual circumstances, professional judgment, or jurisdiction-specific factors that a general synthesis cannot capture. This is a working definition for this paper, not a standard one.

Stakes classification should be a tractable NLP problem, though I have not built or evaluated such a classifier. High-stakes queries have distinct linguistic signatures: they tend to be first-person or second-person framed (‘can I,’ ‘should I,’ ‘what are my rights’), they reference personal situations, and they

occur in identifiable semantic domains. A classifier trained on labeled query data from health, legal, financial, and employment domains may reach high precision on stakes detection without deep semantic understanding, because the domain and framing signals are strong.

5.3 Accuracy-Weighted Re-ranking

When a query is classified as high-stakes, SARI applies an accuracy-weighted re-ranking pass before the LLM receives its source documents. Accuracy-weighted re-ranking adjusts source scores using signals that correlate with accuracy in high-stakes domains rather than engagement: source authority (institutional affiliation, professional credentialing, regulatory recognition), source specificity (does the source address the user's jurisdiction and situation type), and hedging density (does the source appropriately qualify its claims).

The goal of re-ranking is not to eliminate engaging content: it is to ensure that the LLM's synthesis context includes high-accuracy sources in proportion to their actual reliability, rather than in proportion to their engagement performance. In practice, this means authoritative but low-engagement sources (clinical guidelines, jurisdiction-specific legal resources, regulated financial disclosures) receive higher weight in the synthesis context for high-stakes queries than their organic ranking would give them. Formally, this is the direct intervention against Proposition 1: it relaxes the assumption that every member of G strictly dominates every member of A in the ranking signal used for retrieval, by substituting a reliability-correlated score for the pure engagement score on high-stakes queries.

5.4 Stakes-Conditioned Generation

The second intervention point is at LLM generation. SARI introduces stakes-conditioned generation instructions that modify the model's default response behavior for high-stakes queries. Three modifications are applied:

- Uncertainty surfacing: the model is instructed to explicitly represent disagreement and uncertainty across source documents, rather than synthesizing toward a confident consensus. For high-stakes queries, source disagreement is informative: it tells the user that the answer is genuinely uncertain and warrants professional consultation. Formally, this is the direct intervention against Proposition 3: it restores δ_{obs} as a function of Δ_{true} rather than fixing it at 0, so that $p_{verify} = \varphi(\delta_{obs})$ is once again responsive to the sources' actual disagreement. This is the same underlying goal as distribution-free uncertainty quantification methods like conformal prediction (Angelopoulos & Bates, 2021): surfacing calibrated uncertainty rather than suppressing it, applied here at the interface layer rather than the model's output distribution.
- Epistemic humility framing: the model is instructed to frame its response as a summary of available information rather than a definitive answer. This is a framing change, not an accuracy change, but framing may significantly affect user behavior. A response that says 'sources suggest several possibilities; here are the key considerations' would be expected to produce different downstream behavior than one that says 'the answer is X.'
- Professional referral integration: for queries where the stakes exceed what any general synthesis can reliably address, the model is instructed to include a specific, actionable professional referral, not a generic 'consult a professional' disclaimer appended to a confident answer, but a genuine acknowledgment that the question requires individual professional assessment, positioned prominently within the response.

5.5 What SARI Does Not Do

SARI does not solve the hallucination problem. It does not guarantee that LLM search answers are accurate. It does not replace professional judgment for genuinely complex legal, medical, or financial situations. What it does is: preserve some of the epistemic protection that multi-source triangulation previously provided, by ensuring that high-stakes synthesis contexts include accurate sources and that high-stakes responses communicate uncertainty rather than hiding it. The goal is not perfection. It is restoring a protection that the shift to LLM search may have quietly removed.

It is also worth stating precisely what SARI's two interventions do and do not fix, given Propositions 1–3. Accuracy-weighted re-ranking (Section 5.3) targets Proposition 1 by changing which documents enter the synthesis context C; it does not by itself guarantee proportional influence within the synthesis, which is the separate mechanism Proposition 2 describes. Uncertainty surfacing (Section 5.4) targets Proposition 3 directly, but only for disagreement the model can detect within its context, if Proposition 2's dilution has already suppressed a minority accurate source's distinguishability from the majority, there may be little disagreement left for the model to surface. Section 6 found no dilution in either task format I tested, so this interaction may be weaker than the formal model suggests. This is why SARI intervenes at both re-ranking and generation rather than either alone: each stage's fix addresses a different proposition, and neither is sufficient by itself.

6. A Direct Test of Proposition 2: Synthesis Dilution

Proposition 2 is the most directly testable claim in the formal model, because it makes a prediction about a model's output: that a single hedged source loses influence on the synthesis as confident sources are added. I tested it against a live model, in two task formats, using source sets built so that the model could not detect the confident sources just by noticing that they were copies of each other.

6.1 Setup

I used four domains, each with one question: health (is a persistent dry cough after a cold something to worry about), legal (can my employer fire me without warning), financial (should I refinance my mortgage), and hiring (does a job rejection mean discrimination). Each domain has one hedged, accurate source (H) and eight different confident, engagement-style sources (C1 to C8) that I wrote. The confident sources make the same kind of claim in different words, for example "red flag", "act now", or "smart homeowners refinance now". There were six conditions: H alone (H0), H plus 1, 2, 4, or 8 confident sources (k1, k2, k4, k8), and eight confident sources with no hedged source (C8), which shows what the model does when no careful source is present. The position of H in the list was randomized. Each domain and condition was run six times, giving 24 outputs per condition per task.

The generator was Claude Sonnet 4.6 (claude-sonnet-4-6). The two tasks were "Summarize what these sources say about the question" (run October 21, 2026) and "Using the sources above, answer the question for the reader" (run October 30, 2026). That is 288 outputs in total. If the model synthesized like a salience-weighted average with equal weights, the hedged source's share would fall from 1 at H0 to 1/9 at k8, and the stance of the output should move toward the confident claim.

6.2 Scoring and Judge Check

Each output was scored by a rubric judge (also Claude Sonnet 4.6) that saw the source set and the output. It gave a stance from 1 to 5 (1 means the output adopts the overconfident claim, 5 means it is careful and discounts it), said which view the output endorses (hedged, confident, both, or neither), and said whether the output itself tells the reader to act immediately or treats the confident claim as true. To check the judge, I labeled 30 outputs from each task, 60 in all, on the same three questions. I did this blind to

condition and to the judge's scores. The judge's stance was within one point of mine on 60 of 60 outputs and identical on 46 of 60 (77%). It agreed with me on which view the output endorses in 59 of 60, and on urgency in 58 of 60. Where we differed on stance, the judge was usually the more generous one, giving a 5 where I gave a 4.

6.3 Results

Condition	Sources shown	Summarize: mean stance	Answer: mean stance	Overconfident outputs (both tasks)
H0	H only	5.00	4.79	0 of 48
k1	H + 1 confident	4.79	4.96	0 of 48
k2	H + 2 confident	4.92	4.92	0 of 48
k4	H + 4 confident	5.00	4.96	0 of 48
k8	H + 8 confident	5.00	4.92	0 of 48
C8	8 confident, no H	4.83	5.00	0 of 48

No output in 288 was judged to endorse the confident claim. The 95% Wilson upper bound is 1.3% for all 288 outputs and 7.4% for any single condition with its 48 outputs. Of the 288 outputs, 287 scored 4 or 5 on stance, and the lowest was a 3. Stance did not decline as confident sources were added: the k8 means (5.00 and 4.92) are not lower than the H0 means (5.00 and 4.79). In C8, where no hedged source was present, the judge recorded that the output discounted the confident sources in 24 of 24 summaries and 23 of 24 answers, so the model pushed back on sources that agreed with each other. The urgency flag fired on 0 of 144 summaries and 9 of 144 answers. The largest count was in H0 (4), where no confident source exists, which suggests it partly picks up legitimate deadline advice from the hedged source, such as EEOC filing limits. I treat urgency as a secondary measure.

6.4 What This Means

In both task formats, this model did not behave like a salience-weighted average of its sources. The result does not affect the algebra of Proposition 2. It does mean the test gives no support to the claim that a real LLM synthesis behaves that way. Several explanations are possible and I have not separated them. (1) The four topics are widely covered, so the model may be answering from its own knowledge and using the sources as material to evaluate. (2) The confident sources were written in an obviously promotional style, so they may be easy to spot, and subtler confident sources could behave differently. (3) Stance is near its ceiling in every condition, which leaves little room for a decline to show. (4) The judge is from the same model family as the generator. Its agreement with my labels is high, but I am a single rater. (5) With 24 outputs per condition, a true rate of overconfident outputs of up to about 14% in any one condition is not ruled out. (6) I tested one model, up to eight confident sources, and sources I wrote myself.

I do not claim Proposition 2 fails in deployed systems, and I do not claim it holds. I claim that for this model, in these four domains and these two tasks, no dilution appeared. The result is dated and model-specific: a live model changes across versions, so these numbers are not reproducible the way a fixed benchmark is. An earlier pilot that used identical copies of one confident source and an unvalidated keyword count also showed no trend, but its design had weaknesses that the test above fixes, so I do not report it as evidence. Dilution could still appear with weaker priors, such as niche topics, with less obvious confident sources, or with smaller models; those are the next tests. Propositions 1 and 3 do not depend on this result, and neither does the case for SARI's re-ranking (Section 5.3), which rests on Proposition 1, or for its uncertainty surfacing (Section 5.4), which rests on Proposition 3. The prompts, all 288 outputs, the judge scores, and my labels are released with this paper's Zenodo record.

7. Discussion

7.1 The Incentive Problem

I want to be honest about why Triangulation Collapse is hard to fix at the organizational level. The metrics that AI search products optimize for (engagement, satisfaction scores, session length, return visits) all improve when users receive confident synthesized answers. TC is an improvement in product metrics and a deterioration in user outcomes. That gap is structurally difficult to close when the people measuring product success are not the people paying the downstream cost.

There is also a competitive dynamic that makes unilateral correction difficult. An AI search product that introduces uncertainty, hedging, and professional referrals for high-stakes queries will generate lower satisfaction scores than one that delivers confident answers. In a competitive landscape where every product is racing to demonstrate AI capability, restraint looks like weakness. The question of how AI systems should behave when they don't know the answer is being decided by engagement metrics, not by the people whose health and legal and financial situations depend on getting that answer right.

7.2 The Quality Retrospection Gap

Perhaps the deeper issue is that the culture of AI product development does not reward retrospection. Building something new is celebrated. Improving something that already exists, going back to understand where a deployed system is failing, for whom, and in what circumstances, is treated as maintenance work, not innovation. The result is that failure modes like TC can exist for years in deployed systems without being formally identified, because the incentive to identify them is absent.

SARI is not technically complex. A stakes classifier is a standard NLP problem. Accuracy-weighted re-ranking uses signals that are already available in most search infrastructures. Stakes-conditioned generation is a prompt engineering problem. The technical barrier to implementing something like SARI is low. The organizational barrier, convincing a team that their engagement metrics do not capture the full picture of product quality, is much higher.

7.3 Limitations

This paper is primarily a conceptual contribution, and Propositions 1–3 should be read accordingly: they establish that each stage of the decoupling mechanism is structurally capable of producing the described failure under the stated assumptions: strict engagement/reliability dominance for Proposition 1, salience-weighted averaging for Proposition 2, a single-answer interface fixing δ_{obs} at 0 for Proposition 3. They do not establish how often these assumptions hold in a given deployed system, or how large the resulting effects are in practice; that is an empirical question the formal model does not answer on its own. The empirical test I ran (Section 6) found no support for the assumption behind Proposition 2 for one model, in two task formats, and the assumptions behind Propositions 1 and 3 remain untested.

The SARI framework is a design proposal, not a validated system: I have not implemented SARI at scale or measured its effect on user outcomes. The evidence for TC across domains is drawn from argument and from established patterns in the search quality literature, not from a field study of real users. The one controlled experiment I ran tests Proposition 2 only (Synthesis Dilution), on one model, with a judge from the same model family and 30 human labels per task format. It is a real test but a narrow one, and it covers none of the other propositions or the effect of SARI. Future work should include: formal measurement of triangulation behavior change before and after LLM search adoption; empirical validation of SARI's re-ranking and generation components, ideally by directly estimating $\bar{a}_A$, $\bar{a}_G$, and the β_j salience weights in a live system rather than assuming their existence; a repeat of the Synthesis Dilution test with varied rather than duplicated sources, larger n, other domains and models, and human-rated

outputs; and longitudinal studies of user decision quality in high-stakes domains under different search interface conditions.

I also want to acknowledge that the stakes classifier is an imperfect boundary. The line between high-stakes and low-stakes is not always clear, and a classifier will make errors in both directions. Over-triggering SARI for low-stakes queries adds unnecessary friction. Under-triggering it for genuinely high-stakes queries leaves the core problem unaddressed, and by Proposition 1, an under-triggered query is simply routed back into the strict-dominance regime the classifier was meant to break. Calibrating this boundary well is itself a research problem.

8. Conclusion

This paper has described a possible behavioral shift: the replacement of multi-source verification by single-answer synthesis, and a formal mechanism, engagement-accuracy decoupling, that would drive it. The interface presents a single answer as complete, and under the model in Section 3.3 the user's instinct to verify it is structurally suppressed (Proposition 3), provided the interface offers no other view of disagreement. If that change is happening at scale, it applies to millions of users asking questions about their health, their legal rights, and their financial futures.

The propositions are conditional results and not findings about deployed systems. I tested the most testable one, Synthesis Dilution, against a live model and found no dilution in 288 outputs from one model. That result changes how confidently the mechanism can be claimed to explain real behavior: the formal model says what would follow from its assumptions, and the assumption in Proposition 2 found no support in the tests I ran. I report this as a finding of the paper, and as an empirical question this paper opens and does not close.

I am not arguing that LLM search is bad. I am arguing that it may be deployed with an optimization target that was designed for a different problem. Engagement and accuracy correlate well enough for most queries that this mismatch doesn't matter much most of the time. For the subset of queries where someone's health, legal standing, financial security, or employment future is at stake, it matters a great deal. SARI is a first proposal for how to address that mismatch in a practical, deployable way, and Section 5.5 makes explicit which of Propositions 1–3 each of its interventions targets.

The broader point is one about what gets measured. Under the model's assumptions, Triangulation Collapse would be invisible to every standard product metric currently in use, and Proposition 3 shows this is not a gap that better dashboards alone can close, since the interface severs the dependency between user behavior and true source disagreement before the user ever acts. Making that cost visible, and building evaluation infrastructure that can detect it, is the necessary precondition for knowing whether it is large. That is the work I hope this paper motivates.

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Revision note

Version 2 (November 2026) corrects Propositions 1 and 3, removes first-person anecdotes, adds prior work on verifiability and manipulation in generative search, adds a relation to LLM-based evaluation, reports a live-model test of Proposition 2 (a null result across 288 outputs and two task formats), presents the four domains as hypotheses, and reframes the abstract and conclusion accordingly.